$$\left.\begin{aligned} 2x_1x_2 + x_2^3 - x_2 &= 0 \\ x_1^2 + 3x_2^2x_1 - x_1 &= 0 \end{aligned}\right\}$$

hold simultaneously. This means that

$$(x_2 = 0 \text{ or } 2x_1 + x_2^2 = 1) \text{ and } (x_1 = 0 \text{ or } x_1 + 3x_2^2 = 1)$$

i.e. $(x_2 = 0$ and $x_1 = 0)$ or $(x_2 = 0$ and $x_1 + 3x_2^2 = 1)$ or

$(2x_1 + x_2^2 = 1$ and $x_1 = 0)$ or $(2x_1 + x_2^2 = 1$ and $x_1 + 3x_2^2 = 1)$.

The only points at which it is possible for f to have a local maximum or minimum are therefore $(0, 0)$, $(1, 0)$, $(0, 1)$, $(0, -1)$, $(2/5, 1/\sqrt{5})$, $(2/5, -1/\sqrt{5})$. (The final pair of points is obtained by solving

$$\left.\begin{aligned} 2x_1 + x_2^2 &= 1 \\ x_1 + 3x_2^2 &= 1 \end{aligned}\right\}$$

simultaneously.)

It should be emphasised that all that has been shown is that f cannot have a local maximum or minimum at a point other than those listed. It has *not* been shown that f actually does have a local maximum or minimum at any of these points. In fact, f does *not* have a local maximum or minimum at $(0, 0)$. To see this, consider what happens on the lines $x_1 = x_2$ and $x_1 = -x_2$. We have that

$$f(t, t) = t^3 + t^4 - t^2$$

which has a local *maximum* at $t = 0$ while

$$f(t, -t) = t^3 - t^4 + t^2$$

which has a local *minimum* at $t = 0$. The point $(0, 0)$ is an example of a *saddle point* for f. We shall discuss such points in more detail in §19.40.

19.11 Exercise

(1) Let $f\colon \mathbb{R}^2 \to \mathbb{R}$ be defined by $f(x_1, x_2) = x_1^2x_2 + x_2^3x_1$. Compute the directional derivative of f at the point $(1, 2)$ in the direction $(3, 4)$.

(2) Compute the partial derivatives of the functions $f\colon \mathbb{R}^2 \to \mathbb{R}$ defined by

(i) $f(x_1, x_2) = x_1 + 2x_2$ (ii) $f(x_1, x_2) = x_1^2x_2^5$

(iii) $f(x_1, x_2) = \log(x_2 + e^{x_1x_2})^2$ (iv) $f(x_1, x_2) = \cos(x_2 \sin x_1)$.

(3) Find all the points in $\mathbb{R}^2$ at which all partial derivatives of the function $f\colon \mathbb{R}^2 \to \mathbb{R}$ defined by $f(x_1, x_2) = x_1^4x_2 - x_1^2x_2^3$ are zero.

(4) Find all points in $\mathbb{R}^3$ at which all partial derivatives of the function $f\colon \mathbb{R}^3 \to \mathbb{R}$ defined by $f(x_1, x_2, x_3) = x_1x_2x_3$ are zero. Show that none of these points corresponds to a local maximum or minimum.

(5) The *component functions* f^i: $\mathbb{R}^n \to \mathbb{R}$ of a function f: $\mathbb{R}^n \to \mathbb{R}^m$ are defined by

$$f(\mathbf{x}) = (f^1(\mathbf{x}), f^2(\mathbf{x}), \ldots, f^m(\mathbf{x})).$$

Explain why

$$D_{\mathbf{u}}f(\boldsymbol{\xi}) = (D_{\mathbf{u}}f^1(\boldsymbol{\xi}), D_{\mathbf{u}}f^2(\boldsymbol{\xi}), \ldots, D_{\mathbf{u}}f^m(\boldsymbol{\xi})).$$

(6) Prove that the function f: $\mathbb{R}^2 \to \mathbb{R}$ defined by

$$f(x_1, x_2) = \begin{cases} \dfrac{x_1^2 x_2}{x_1^4 + x_2^2} & (x_1, x_2) \neq (0, 0) \\ 0 & (x_1, x_2) = (0, 0) \end{cases}$$

has directional derivatives in every direction at (0, 0) but is *not* continuous at (0, 0). [Hint: example 18.43.] Does f have a local maximum or minimum at (0, 0)?

19.12 Differentiable functions

Consider a function f: $A \to \mathbb{R}^m$ where A is a set which contains each $\mathbf{x} \in \mathbb{R}^n$ satisfying $\|\mathbf{x} - \boldsymbol{\xi}\| < \Delta$. What should it mean to say that f is differentiable at $\boldsymbol{\xi}$? We dealt with the case $m = n = 1$ in §10.1. Our definition required the existence of the limit

$$l = \lim_{x \to \xi} \frac{f(x) - f(\xi)}{x - \xi} = \lim_{h \to 0} \frac{f(\xi + h) - f(\xi)}{h}.$$

This may be rewritten as

$$f(x) - f(\xi) - l(x - \xi) = o(x - \xi) \qquad (x \to \xi)$$

or as

$$f(\xi + h) - f(\xi) - lh = o(h) \qquad (h \to 0)$$

where $o(h)$ denotes a quantity which tends to zero when divided by h. The number l, of course, is called the derivative of f at ξ and denoted by $f'(\xi)$.

An affine function g: $\mathbb{R} \to \mathbb{R}$ is defined by an equation of the form $g(x) = lx + c$. (See example 18.24(i).) We can therefore restate the definition yet again as being the requirement that there exist an affine function g: $\mathbb{R} \to \mathbb{R}$ with $g(\xi) = f(\xi)$ such that

$$f(x) - g(x) = o(x - \xi) \qquad (x \to \xi)$$

The geometric interpretation of this condition is that the graphs $y = f(x)$ and $y = g(x)$ are tangent where $x = \xi$.